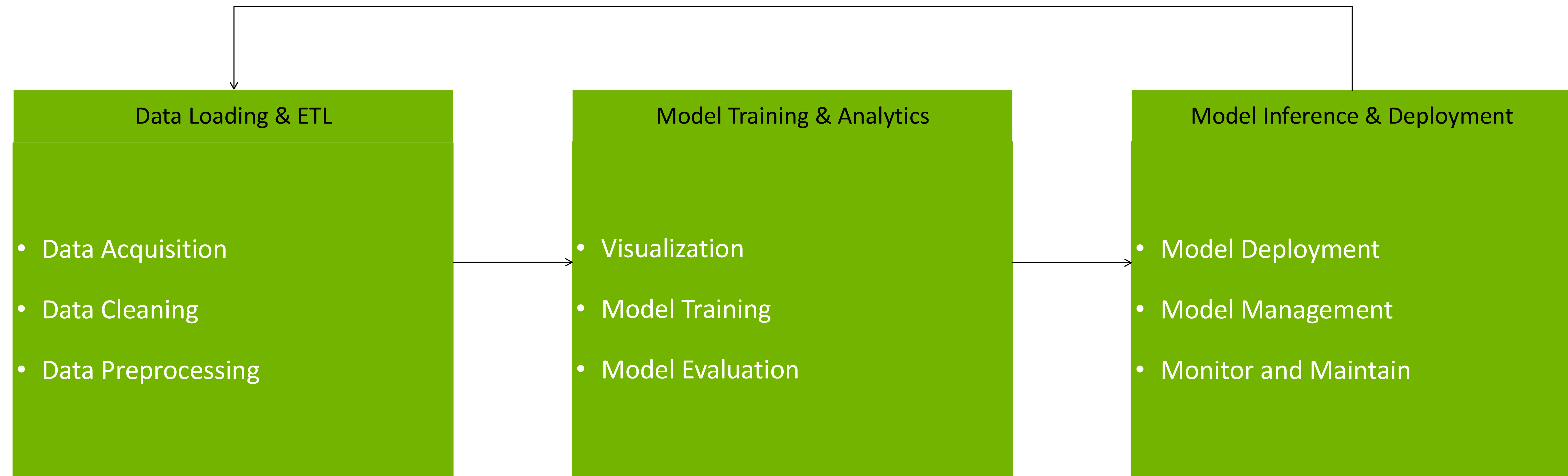




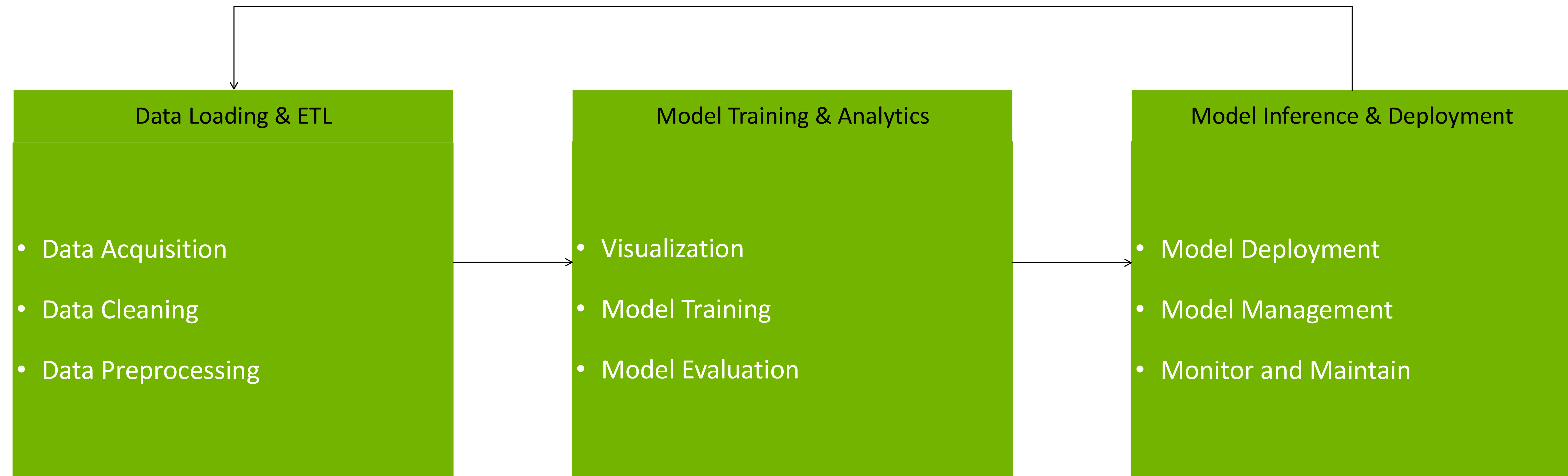
Accelerate Data Science Workflows with Zero Code Changes

NVIDIA Deep Learning Institute

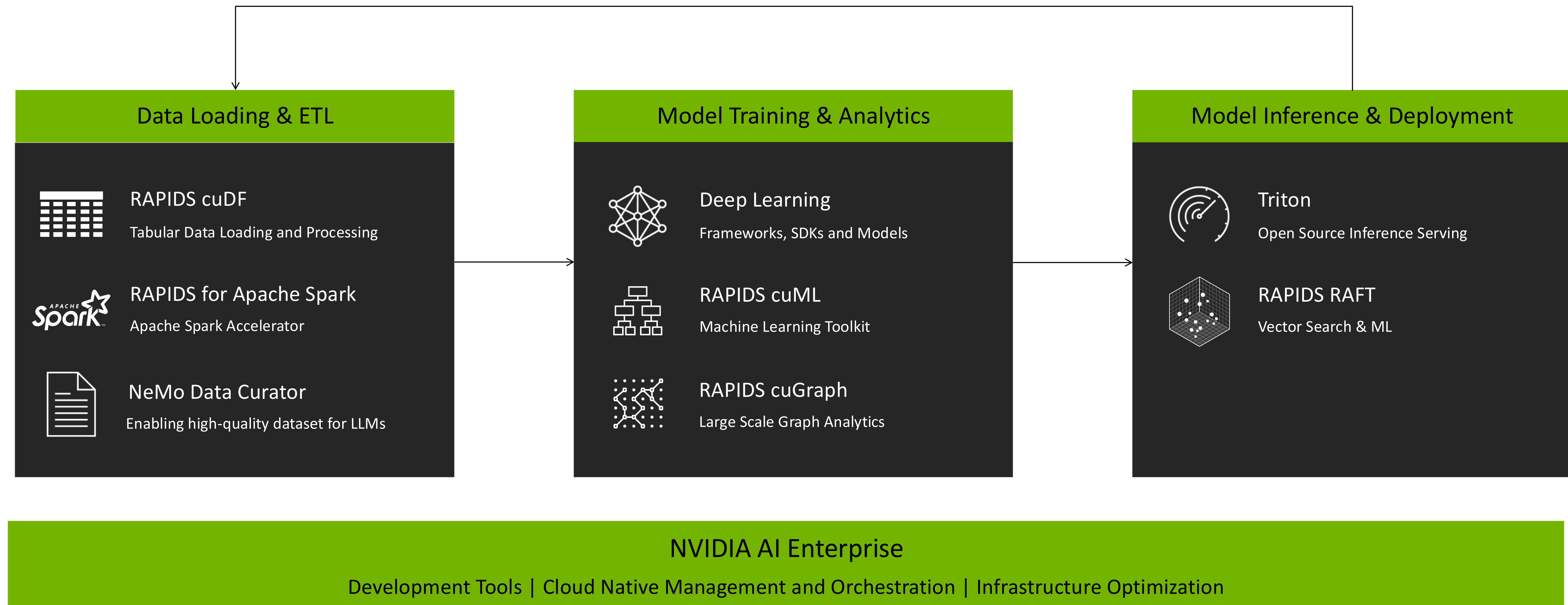
Data Science Workflow



Data Science Tools



Data Science Can Be Accelerated End-to-End Today



Adoption to New Tools Can Be a Challenge

- API Coverage
 - Problem: learning new APIs takes time
- Compatibility
 - Problem: using new tools for each stage of the workflow may adversely impact downstream processes
- Specific Hardware Availability
 - Problem: hardware-acceleration requires the specific hardware for testing and development
- Solution: RAPIDS integrates with libraries within the PyData ecosystem to provide acceleration with zero code change

Accessibility to GPU-Acceleration via RAPIDS

- Over time, RAPIDS will become more available as it seamlessly integrates with the ecosystem
- Three specific features to highlight:
 - cuDF
 - cuML
 - cuGraph

cuDF pandas accelerator mode (cudf.pandas)

Bringing world-class performance to pandas workflows

- **Zero Code Change Acceleration**

- Write your code with the full flexibility of pandas. Just load *cudf.pandas* to accelerate on the GPU, with automatic CPU fallback if needed.

- **Third-Party Library Compatible**

- Pass Pandas objects to third-party libraries in cudf.pandas mode and things just work. It will even accelerate pandas operations within these libraries.

- **One Code Path**

- Develop, test, and run in production with a single code path, regardless of hardware.

- **Designed to Accelerate the Entire pandas Ecosystem**

- Bringing the speed of cuDF to pandas users and the ecosystem built for them.

Notebook/IPython:

```
%load_ext cudf.pandas
```

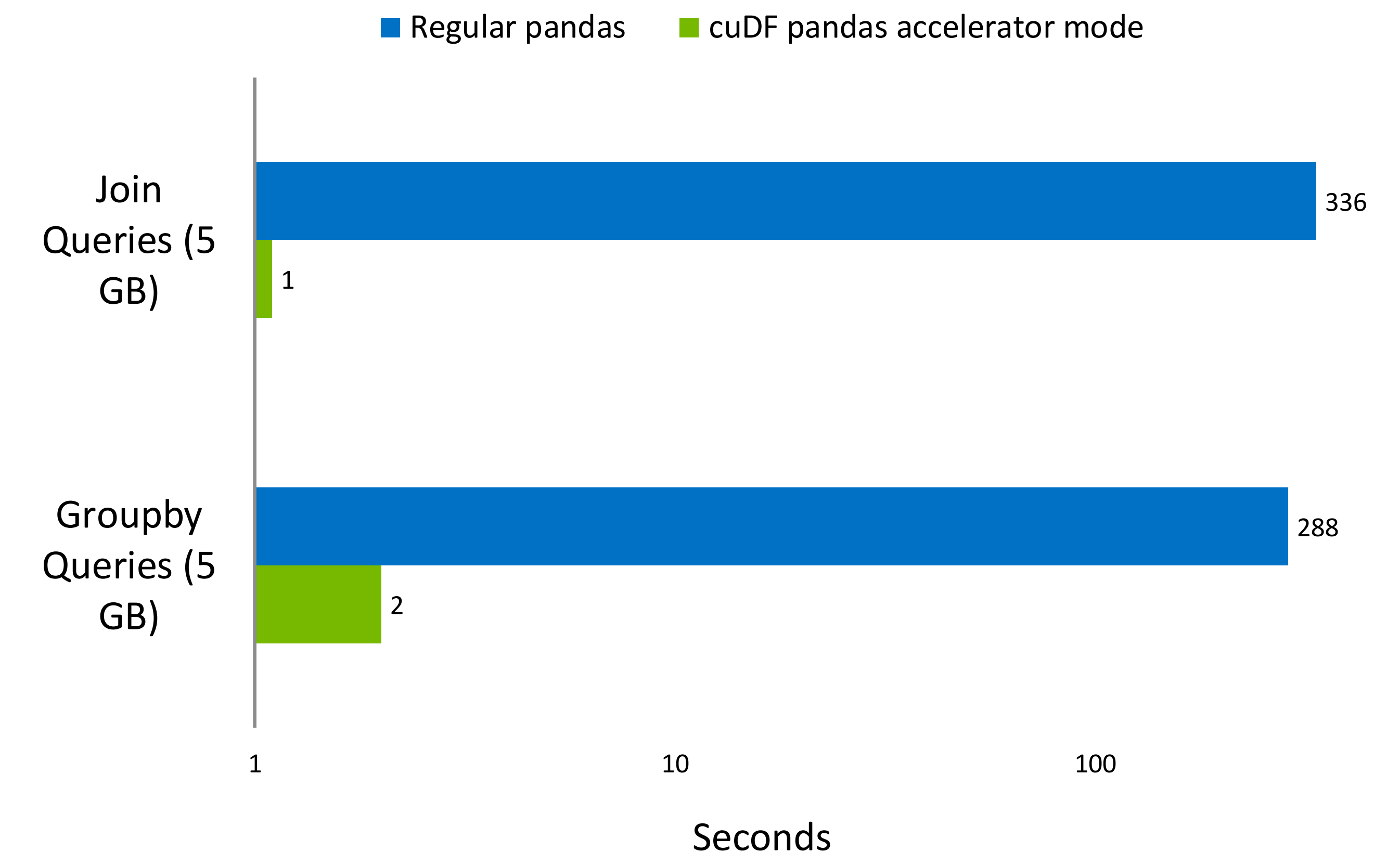
Command-line:

```
python -m cudf.pandas <args>
```

Import:

```
import cudf.pandas
cudf.pandas.install()
```

Standard DuckDB Database-like Ops Benchmark (5GB)



HW: NVIDIA Grace Hopper, CPU: Intel Xeon Platinum 8480CL
SW: pandas v1.5, RAPIDS cuDF 23.10

The cuGraph Backend for NetworkX

Bringing GPU-accelerated graph analytics to every NetworkX user

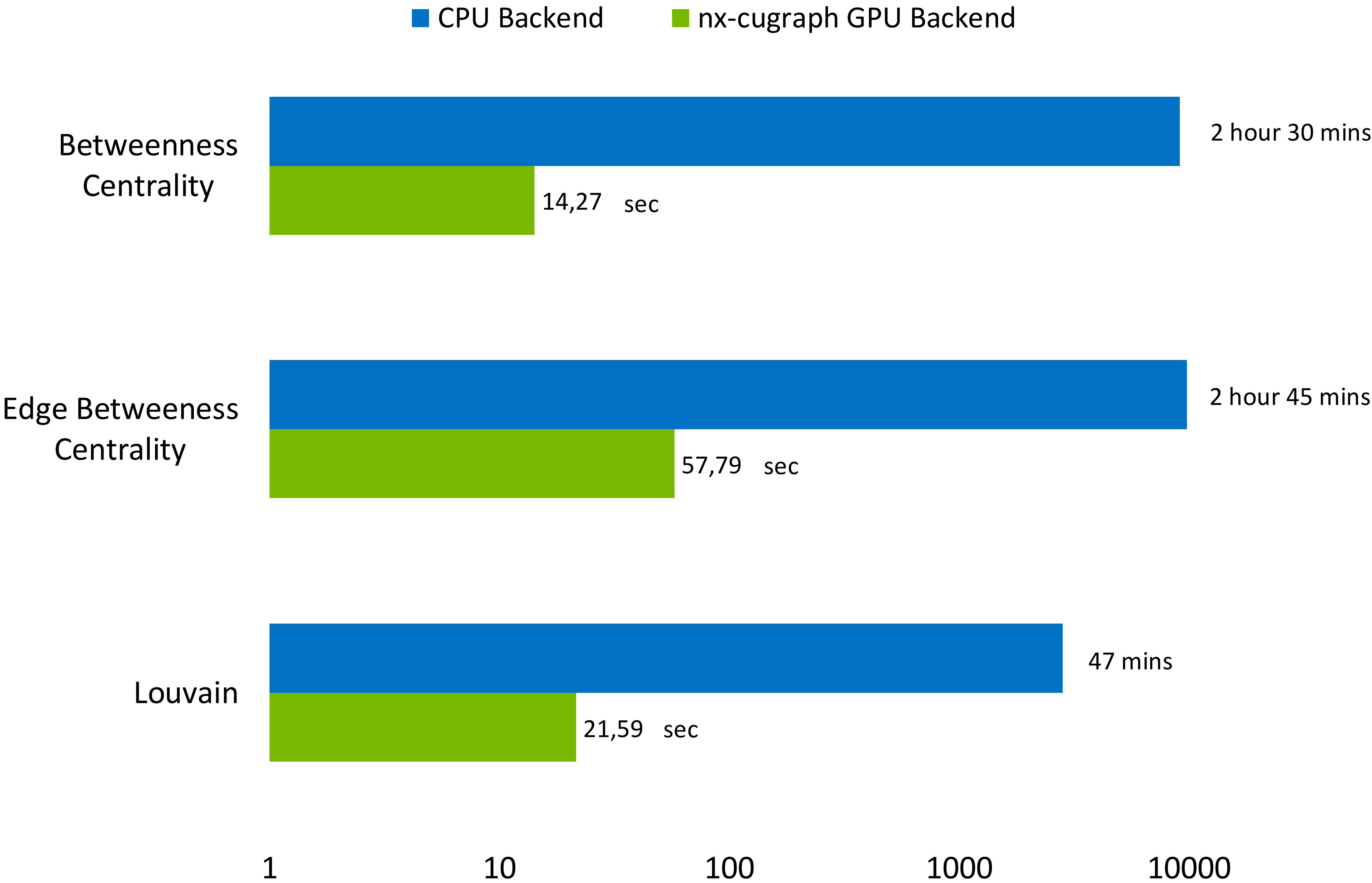
- **Zero Code Change Acceleration**
 - Just configure NetworkX to use the cuGraph backend through environment variables
- **One Code Path**
 - Develop, test, and run in production with a single code path, regardless of hardware.



NetworkX
Network Analysis in Python

+

RAPIDS

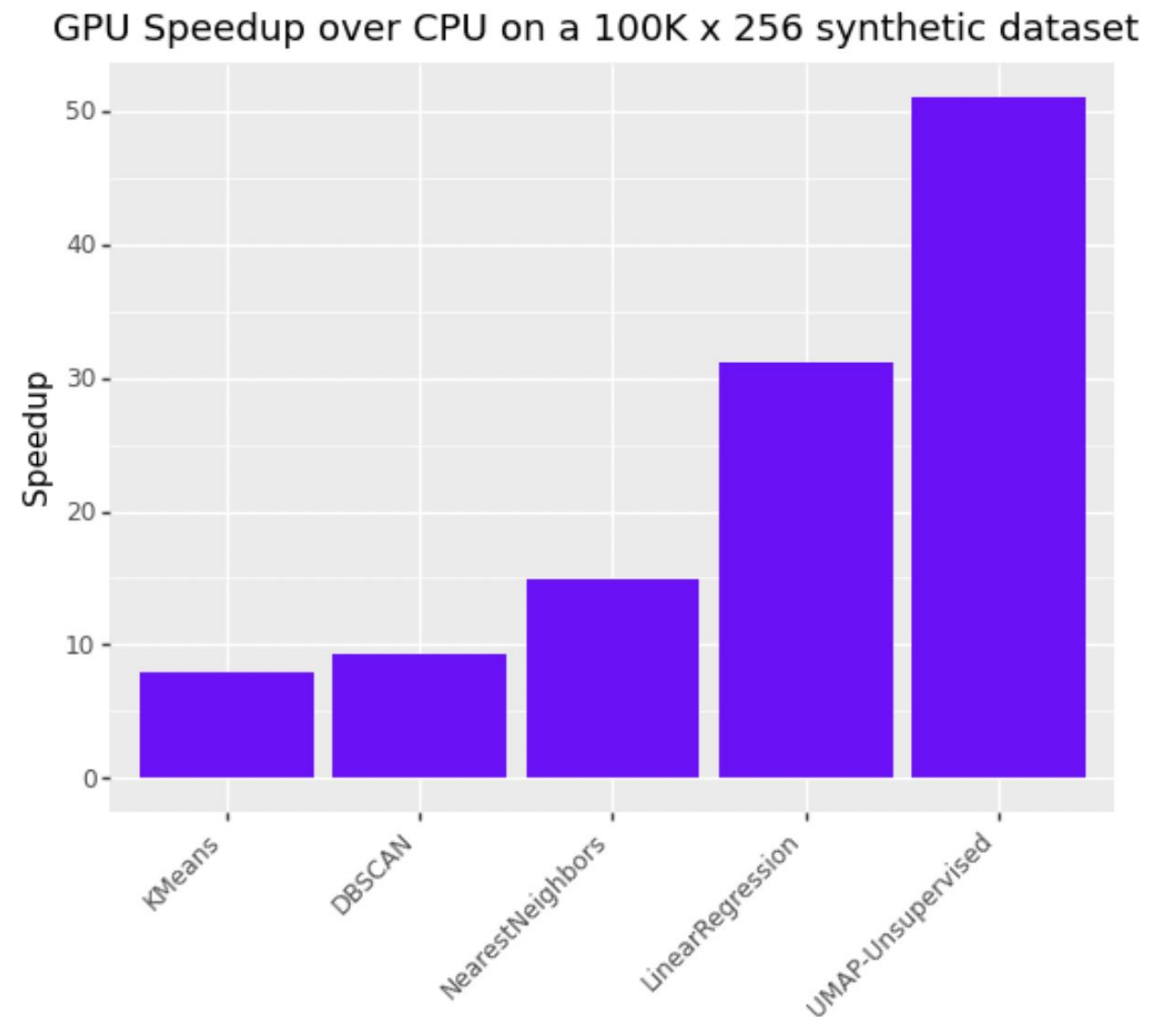
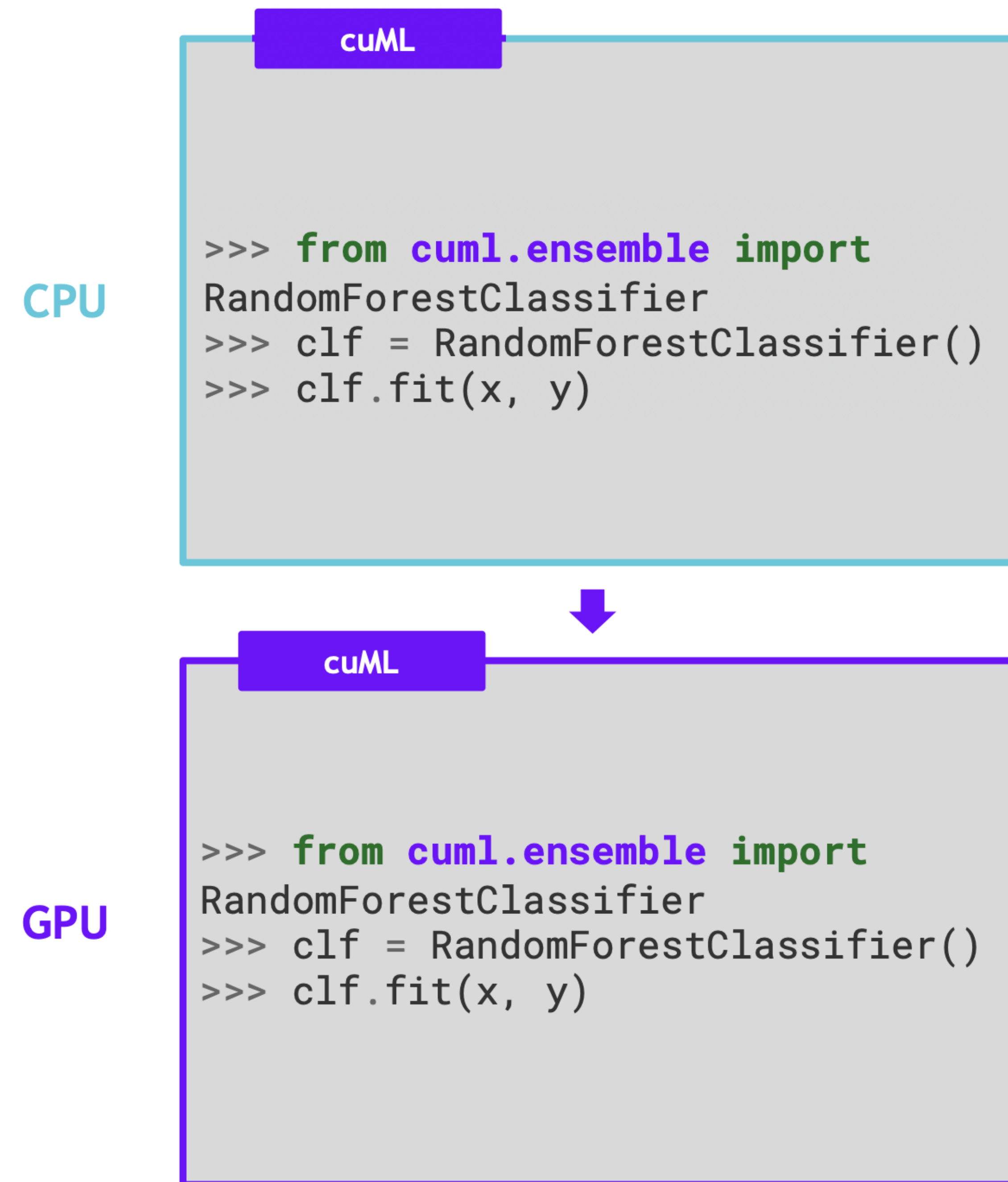


nx-cugraph on NVIDIA H100 vs. NetworkX on Intel Xeon Platinum 8480CL CPU
Dataset: US Patents hosted by Stanford Snap

Up to 600X Faster

A Unified CPU and GPU cuML Experience

Making accelerated machine learning accessible to every data scientist



NVIDIA H100 vs. Intel Xeon Platinum 8480CL
cuML Python vs. Scikit-learn

